

UNIT 10: THE SEARCH AND RESCUE SYSTEM

I. INTRODUCTION

Search and rescue at sea is an activity that demonstrates the noble humanitarian spirit of all organizations and individuals, as well as the obligations and responsibilities of all countries, especially for maritime countries. The following lesson will introduce the global search and rescue organization system and the basic definitions of search and rescue.



II. TEXT

A. The Search and Rescue System (SAR)

1. Global SAR System Organization

The International Civil Aviation Organization (ICAO) and the International Maritime Organization (IMO) coordinate, on a global basis, member States' efforts to provide search and rescue (SAR) services. Briefly, the goal of ICAO and IMO is to provide an effective world-wide system, so that wherever people sail or fly, SAR services will be available if needed. The overall approach a State takes in establishing, providing, and improving SAR services is affected by the fact that these efforts are an integral part of a global SAR system.

A basic, practical, and humanitarian effect of having a global SAR system is that it eliminates the need for each State to provide SAR services for its own citizens wherever they travel world-wide. Instead, the globe is divided into search and rescue regions (SRRs), each with a rescue co-ordination centre (RCC) and associated SAR services, which assist anyone in distress within the SRR without regard to nationality or circumstances.

2. National and Regional SAR System Organization

States, by being Party to the Safety of Life at Sea (SOLAS) Convention, the International Convention on Maritime Search and Rescue, and the Convention on International Civil Aviation, have accepted the obligation to provide aeronautical and maritime SAR co-ordination and services for their territories, territorial seas, and, where appropriate, the high seas. SAR services are to be available on a 24-hour basis. To carry out these responsibilities, a State either should establish a national SAR organization, or join one or more other States to form a regional SAR organization.

The SAR system has three levels of co-ordination associated with SAR coordinators (SCs), SAR mission coordinators (SMCs), and on-scene coordinators (OSCs).

SAR Coordinators (SCs): SCs have the overall responsibility for establishing, staffing, equipping, and managing the SAR system, including providing appropriate legal and funding support, establishing RCCs and rescue sub-centres (RSCs), providing or arranging for SAR facilities, coordinating SAR training, and developing SAR policies. SCs are the top level SAR managers; each State normally will have one or more persons or agencies for whom this designation may be appropriate. SCs are not normally involved in the conduct of SAR operations

SAR Mission Coordinator (SMC): Each SAR operation is carried out under the guidance of an SMC. This function exists only for the duration of a specific SAR incident and is normally performed by the RCC chief or a designee. For complex cases or those of long duration, the SMC usually has an assisting team. The SMC is in charge of a SAR operation until a rescue has been effected or until it has

become apparent that further efforts would be of no avail, or until responsibility is accepted by another RCC. The SMC should be able to use readily available facilities and to request additional ones during the operation. The SMC plans the search and coordinates the transit of SAR facilities to the scene. The SMC should be well trained in all SAR processes and be thoroughly familiar with the applicable SAR plans. The SMC must competently gather information about distress situations, develop accurate and workable action plans, and dispatch and coordinate



the resources which will carry out SAR missions. The plans of operation maintained by the RCC provide information to assist in these efforts.

Search and rescue drills at sea

3. SAR Co-ordination

On-scene Coordinator (OSC): When two or more SAR units are working together on the same mission, there is sometimes an advantage if one person is assigned to coordinate the activities of all participating units. The SMC designates

this on-scene coordinator (OSC), who may be the person in charge of a search and rescue unit (SRU), ship or aircraft participating in a search, or someone at another nearby facility in a position to handle OSC duties. The person in charge of the first SAR facility to arrive at the scene will normally assume the function of OSC until the SMC directs that the person be relieved. The OSC should be the most capable person available, taking into consideration SAR training, communications capabilities, and the length of time that the unit the OSC is aboard can stay in the search area. Duties which the SMC may assign to the OSC, depending on needs and qualification, include any of the following:

- Assume operational co-ordination of all SAR facilities on-scene;
- Receive the search action plan from the SMC;
- Modify the search action plan based on prevailing environmental conditions and keeping the SMC;
- Advised of any changes to the plan (do in consultation with the SMC when practicable);
- Provide relevant information to the other SAR facilities;
- Implement the search action plan;
- Monitor the performance of other units participating in the search;
- Co-ordinate safety of flight issues for SAR aircraft;
- Develop and implement the rescue plan (when needed); and make consolidated reports (SITREPs) back to the SMC

4. SAR Resources

The SAR organization includes all of those agencies which perform distress monitoring, communications, co-ordination, and response functions. This includes providing or arranging for medical advice, initial medical assistance, or medical evacuation, if necessary. SAR facilities consist of all of the public and private facilities, including co-operating aircraft, vessels, other craft and installations operating under co-ordination of an RCC. In establishing a SAR service, States should use existing facilities to the fullest extent possible. A

successful SAR organization usually can be created without having designated, full time SRUs.

Ship Reporting System: Vessels at sea, although not always available to participate in extended search operations, are potential SAR assets. Masters of vessels have a duty to assist others whenever it can be done without endangering the assisting vessel or crew. Various States have implemented ship reporting systems. A ship reporting system enables the SMC to quickly know the approximate positions, courses, and speeds of vessels in the vicinity of a distress situation by means of a surface picture (SURPIC), and other information about the vessels which may be valuable.

The Amver system: The only world-wide system operated exclusively to support SAR, makes information available to all RCCs. Any United States RCC can be contacted for this type of SAR information.

Global Maritime Distress and Safety System: After 31 January 1999, ships subject to the SOLAS Convention should be outfitted with certain communications equipment, collectively referred to as the shipboard portion of the Global Maritime Distress and Safety System (GMDSS). Certain fishing vessels and



other marine craft also may be obligated to carry GMDSS-compatible equipment, or may do so voluntarily. GMDSS is intended to provide automatic alerting and locating with minimal delay, a reliable network for SAR communications, integration of satellite and terrestrial communications, and adequate frequencies in all maritime bands.

First response in search and rescue

5. SAR Operations Stages

The success of a SAR mission often depends on the speed with which the operation is planned and carried out. The prompt receipt of all available information by the RCC is necessary for thorough evaluation of the situation, immediate decision on the best course of action, and a timely activation of SAR facilities. While no two SAR operations follow exactly the same pattern, SAR incidents do generally pass through defined stages, which can be used to help organize response activities. These stages are discussed in general terms below and expanded discussion is found in the remaining chapters of this Volume. These stages should be interpreted with flexibility, as many of the actions described may be performed simultaneously or in a different order to suit specific circumstances.

Awareness Stage: A period during which the SAR system becomes aware of an actual or potential incident. The SAR organization cannot respond to an incident until it becomes aware that people or craft need assistance. Therefore, the general public should be encouraged to report any abnormal occurrence which they have heard about or witnessed. SAR authorities must ensure that notification that an aircraft has crashed, or that an aircraft, ship or other craft is overdue or in a state of emergency, can reach an RCC from any source, either directly or via an alerting post. A coast radio station (CRS) usually receives the first information that a ship or other craft on the water is in distress. A CRS is required by international regulations to relay this information to SAR authorities. As a result, an RCC will often receive first notification that a ship or other craft is in distress from a CRS with which it is associated, or via its own communications facilities. The RCC must keep a complete record of information it receives. Pre-

printed forms often are used to ensure that full information about the SAR incident is obtained and remains available for review.

Initial Action: Initial Action stage is a period during which preliminary action is taken to alert SAR facilities and obtain amplifying information. Once an RCC receives an initial report about persons or craft in distress, some immediate action often is appropriate pending receipt and evaluation of more complete information. RCCs usually have in their plans of operation a checklist of steps to accomplish for each type of incident with which the RCC expects that it may become involved. After evaluating all available information and taking into account the degree of emergency, the SMC should declare the appropriate emergency phase and immediately inform all appropriate centers, personnel and facilities. Three emergency phases have been established for classifying incidents and to help in determining the actions to be taken for each incident. These are: uncertainty phase; alert phase; and distress phase.

Planning Stage: Planning Stage is a period during a SAR incident when an effective plan of operations is developed. Comprehensive planning of SAR response tasks is essential, especially when the location of the distress situation is unknown and the survivors move due to wind and water currents. Proper and accurate planning is critical to SAR mission success; if the wrong area is searched, there is no hope that search personnel will find the survivors, regardless of the quality of their search techniques or the amount of their search effort. This requires proper training of the SMC and other RCC watchstanders. Computers can eliminate much of the detailed work in search planning, and can improve accuracy.

Operations Stage: The SAR operations stage encompasses all activities that involve searching for the distressed persons or craft, providing assistance, and removing them to a safe place. In this stage, the SMC assumes a monitoring and guidance role, ensuring that the search plan is received, understood, and followed by SAR facilities. The RCC staff usually will spend most of this stage planning subsequent searches, based on updated information and the assumption that the present search will be unsuccessful.

Conclusion Stage: Conclusion Stage is a period during a SAR incident

when SAR facilities return to their regular location and prepare for another mission. SAR operations enter the conclusion stage when:

- Information is received that the aircraft, ship or persons who are the subject of the SAR incident are not in distress;

- The aircraft, ship or persons for whom SAR facilities are searching have been located and the survivors rescued;

- During the distress phase, the SMC determines that further search would be to no avail because the area has been adequately searched and all probability areas have been investigated or because there is no longer any reasonable probability of survival of the persons on board.

****Distress signals targeted to anyone nearby***

Most important to your operation are the distress signals that you may see or hear. In some situations, you may be the only link between the vessel in distress and the SAR system. Knowing the distress signals and knowing how to respond to them is of paramount importance for anyone involved in maritime search and rescue.

These include the following types:



RED STAR
SHELLS



FOG HORN
CONTINUOUS
SOUNDING



FLAMES ON
A VESSEL



GUN
FIRED AT
INTERVALS OF
1 MIN.

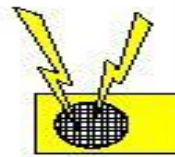


ORANGE
BACKGROUND
BLACK BALL
AND SQUARE

SOS



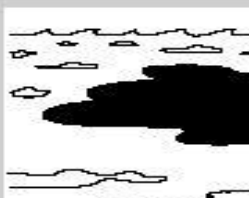
SOS



"MAYDAY"
BY RADIO



PARACHUTE
RED FLARE



DYE
MARKER
(ANY COLOR)



CODE FLAGS
NOVEMBER
CHARLIE



SQUARE FLAG
AND BALL



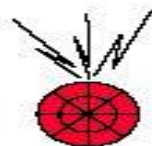
WAVE
ARMS



RADIO-
TELEGRAPH
ALARM



RADIO-
TELEPHONE
ALARM



POSITION
INDICATING
RADIO
BEACON



SMOKE

III. GRAMMAR

1. Possessive noun "'s"

1.1. Use

- Possessive case is mainly used for people or animals, for time, expressions, distance...

E.g. teacher's book, Nam's toys....

- Note that when the possessive case is used, the article before the person or thing "possessed" disappears.

Eg. The daughter of my uncle = My uncle's daughter.

The letter of my girlfriend = My girlfriend's letter.

- Do not use possessive case for thing, idea... We use "of" with thing or idea.

Eg. The daughter of my uncle = My uncle's daughter.

The foot of a tree

The back of the car

1.2. Form

a) Singular noun

A singular noun + 's

Eg. Hoa's book

Mr Ha's child

tomorrow's lesson

yesterday's meeting

this week's holiday

today's lesson

b) Plural not ending in "s"

A plural not ending in s + 's

Eg. The children's toys

c) Plural ending in "s"

A plural ending in s +'

Eg. some teachers' book

2. Present Continuous tense: positive and negative

2.1. Use

- The Present Continuous tense denotes an action happening at the moment of speaking.

E.g. We are learning English now.

- It also denotes an action happening around the moment of speaking. E.g. My mother is watching "The eight year old bride" these days. - Sometimes, it denotes an action at a certain time in the future.

E.g.

I am coming to a party tonight.

Note: Some adverbs of times are used in this tense: now, at the moment, at this time, at present, right now.

2.2. Forms

a) *Positive:*

S+ be (am/is/are) + V-ing...

- Most verbs just add -ing

E.g. wear - wearing

cook - cooking

go-going

learn - learning

- If the verb ends in -e, drop-e before adding -ing

E.g. write - writing

come – coming

- When one syllable verb has one vowel and ends in a conso, double the consonant.

E.g. sit - sitting

get – getting

run - running

swim - swimming

b) Negative:

S+am/is/are + not + V-ing...

3. Articles: A/an

3.1. Use

- "A" is one of the article used before singular countable nouns that begin with a consonant sound.

E.g. a **b**ag /ə **b**æg/
 a **s**tudent /ə 'stju:dnt/

- "An" is used before singular countable nouns that begin with a vowel sound.

E.g. an **a**pple /ən 'æpl/
 an **u**mbrella /ən **ʌ**m'brelə/

3.2. Form

A/AN+ countable and singular noun

But: a unit /ə 'ju:nit/
 a university /ə ju:nɪ'vɜ:sti/
 a uniform /ə 'ju:nɪfɔ:m/

IV. VOCABULARY

1	SAR (Search and Rescue)	/sɜ:rtʃ ənd 'reskju:/'	n	Tìm kiếm và cứu nạn
---	-------------------------	------------------------	---	---------------------

2	ICAO (The International Civil Aviation Organization)			Tổ chức hàng không dân dụng quốc tế
3	IMO (The International Maritime Organization)			tổ chức hàng hải quốc tế
4	integral	/ˈɪntɪgrəl/	a	Tính thiết thực, cần thiết
5	humanitarian	/hjuːˌmænɪˈterɪən/	a	Nhân đạo
6	eliminate	/ɪˈlɪmɪneɪt/	v	Loại trừ
7	SRR (search and rescue regions)	/sɜːrtʃ ənd ˈreskjʊːˈrɪːdʒən/	n	Khu vực tìm kiếm và cứu nạn
8	aeronautical	/erəˈnɔːtɪkl/	a	Thuộc về hàng không
9	designation	/ˌdeɪzɪɡˈneɪʃn/	n	Sự chỉ định
10	thoroughly	/ˈθɜːrəli/	adv	Hoàn toàn, thấy đáo
11	competently	/ˈkɑːmpɪtəntli/	adv	Thành thạo, điều luyện
12	on-scene coordinator (OSC)		n	Điều phối viên hiện trường
13	Modify	/ˈmɑːdɪfaɪ/	v	Làm bớt, làm thay đổi

14	Implement	/ˈɪmplɪment/	v	Công cụ, phương tiện
15	evacuation	/ɪˌvækjuˈeɪʃn/	n	Di tản
16	potential	/pəˈtenʃl/	a	Tiềm tàng, tiềm năng

V. PRACTICE

Exercise 1. Answer these questions

- Explain the chain of command in SAR system
- Who has authority to designate On-Scene Coordinator?
- What criteria does the SMC need to consider before designating OSC?

Exercise 2. Chose the correct answer

- The (soldier/soldier's) uniform is neatly pressed.
- The (general/general's) orders were clear.
- The (sergeant/sergeants') squad was ready for inspection.
- The (captain/captains') ship was at sea.
- The (admiral/admirals') fleet was in formation.
- The (colonel/colonels') regiment was on the march.
- The (lieutenant/lieutenants') platoon was on patrol.
- The (major/majors') battalion was in the field.
- The (commander/commanders') unit was on a mission.
- The (officer/officers') rank was clearly displayed.

Exercise 3. Chose the correct answer

- The (tank/tanks') cannon was aimed at the target.

2. The (helicopter/helicopters') rotors were spinning.
3. The (jet/jets') engines were roaring.
4. The (submarine/submarines') periscope was raised.
5. The (missile/missiles') warhead was armed.
6. The (rifle/rifles') barrel was clean.
7. The (grenade/grenades') pin was pulled.
8. The (pistol/pistols') magazine was loaded.
9. The (bullet/bullets') casing was ejected.
10. The (bomb/bombs') fuse was lit.

Exercise 4. Fill the gap with missing words

1. The soldier carried _____ rifle on his shoulder.
2. The general gave _____ order to his troops.
3. The sergeant led _____ squad of soldiers.
4. The captain commanded _____ ship at sea.
5. The admiral oversaw _____ entire fleet.
6. The colonel led _____ regiment of soldiers.
7. The lieutenant was in charge of _____ platoon.

Exercise 5. Fill the gap with missing words

1. The tank fired _____ shell at the target.
2. The helicopter landed on _____ aircraft carrier.
3. The jet flew at _____ incredible speed.
4. The submarine dove to _____ great depth.
5. The missile was launched from _____ underground silo.
6. The rifle fired _____ bullet at the target.

7. The grenade exploded with _____ loud bang.

Exercise 6. Work in pairs. Tell your partner about something that you are (or have been)

1. successful in.

5. pleased with.

2. excited about.

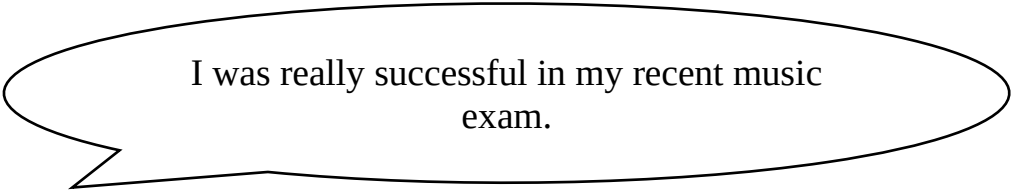
6. good at.

3. satisfied with.

7. angry about.

4. curious about.

8 keen on.



I was really successful in my recent music exam.